

Zen Nguyen

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TECHNICAL SKILLS

- **Languages:** Python, Go, Java, C, C++, SQL
- **Backend & Systems:** RESTful APIs, FastAPI, Node.js, Express, gRPC, TCP Sockets
- **Infrastructure & DB:** SQL, PostgreSQL, MongoDB, Git, Docker, AWS (EC2)
- **AI & Data:** LangChain, OpenAI API, Anthropic API, RAG Architecture, Vector Databases (FAISS/ChromaDB)

TECHNICAL PROJECTS

TitanSwarm (Autonomous AI Co-Pilot) & TitanStore (Raft Database) Jan 2026 – Present

Personal Project · Python, Go, LangChain, FAISS

- Optimized job application workflows by architecting an autonomous AI Co-Pilot in Python, resulting in the automated discovery and parsing of thousands of postings.
- Ensured zero-hallucination data delivery by building a RAG engine using LangChain and OpenAI APIs, leveraging **AI-assisted coding tools** to deliver ATS-optimized resumes.

Gridlock Casino (2D Arcade Engine) Feb 2026 – Present

Collaborative Project · Java, Swing, Maven, JUnit

- Achieved 60-FPS rendering and stable game state management by architecting a custom 2D grid-based engine in Java using a strict MVC architecture.
- Enhanced autonomous enemy behavior by implementing BFS pathfinding and hitscan vector math, resulting in precise projectile collision and visibility.
- Reduced team synchronization errors by managing build automation and dependency resolution via Apache Maven for a 6-developer team.
- Increased engine reliability by validating core logic through comprehensive JUnit testing, ensuring the ability to **troubleshoot and resolve software issues** effectively.

SFU Course Tracker (sfucourseplanner.me) Nov 2025 – Jan 2026

Personal Project · Python, Docker, AWS, FastAPI

- Reduced infrastructure costs and optimized deployments by migrating the full-stack platform from Azure to AWS (EC2) using Docker container orchestration.
- Improved data retrieval for complex queries by designing a normalized **SQL** schema with JSON-type columns to store recursive tree structures.
- Increased data throughput by 10x while maintaining server stability by engineering a scraping pipeline using Asyncio and HTTPX with semaphore-based rate limiting.
- Enabled seamless integration and data exchange by developing **RESTful APIs** via FastAPI to tokenize and evaluate nested boolean prerequisite strings.

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Personal Project · config, github-config

- Maintained profile consistency by managing GitHub configuration files using **Git** for version control.
- Streamlined profile presentation by organizing repository settings and configuration data.

EDUCATION

Bachelor of Science, Computing Science May 2025 – Present

Simon Fraser University

- CGPA: 3.74 / 4.33
- Relevant Coursework: Discrete Math (A+), Computer Systems(A), Data Structures & Programming (B+), Linear Algebra(A-).
- Langara College | Associate of Science, Computer Science Jan 2024 – Apr 2025

WORK EXPERIENCE

Server · Pho Goodness Restaurant Jan 2024 – Present

Burnaby, BC

- Enhanced customer satisfaction for 100+ guests per shift by coordinating effectively with diverse teams under pressure, demonstrating the communication skills required for **technical support**.
- Maintained high academic performance (3.74 CGPA) while working 20+ hours/week, proving the ability to manage project timelines and prioritize tasks in a fast-paced environment.